Since V is indecomposable, one of these summands has to be zero. If the first summand is zero, then φ has to be surjective. If the second summand is zero, then the first one has to be of the desired form, because else we could write it as a direct sum of several objects of the type

$$\begin{tikzcd} & 1 & \\ 0 \arrow[r] & \bullet & 0 \arrow[l] \\ & 0 \arrow[u] & \end{tikzcd}$$

which is impossible since V was supposed to be indecomposable.

(2) This follows similarly by splitting away the kernel of ψ. $\square$

Proposition 6.6.6. *Let Q be a quiver, and let V be a representation of Q.*

(1) If

$$\varphi : \bigoplus_{j \to i} V_j \to V_i$$

is surjective, then

$$F_i^- F_i^+ V = V.$$

(2) If

$$\psi : V_i \to \bigoplus_{i \to j} V_j$$

is injective, then

$$F_i^+ F_i^- V = V.$$

Proof. In the following proof, by $i \to j$ we will always mean that i points into j in the original quiver Q. We only establish the first statement and we also restrict ourselves to showing that the spaces of V and $F_i^- F_i^+ V$ are the same. It is enough to do so for the ith space. Let

$$\varphi : \bigoplus_{j \to i} V_j \to V_i$$

be surjective and let

$$K = \ker \varphi.$$